

Ian Samuel Jackson

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Interests

Hardware Design: Physical design, Computer architecture, CPU/GPU design, FPGAs

Experience

Apple Cupertino, CA
SoC Physical Design Intern May 2023 – Aug. 2023 | May 2024 – Present

- Developed and integrated an internal tool to address timing violations and congestion issues, improving the quality of the design by at least 30% using liner programming techniques.
- Engineered an internal power debugging/analysis tool using TCL and Python to optimize chip designs.
- Implement the physical design block using PnR EDA tools.

ManTech Clarksburg, WV
Application Developer Intern Feb. 2022 – Apr. 2023

- Programmed automated test on FBI's Crime Data Explorer (CDE), Uniform Crime Reporting (UCR), and Use-of-Force (UOF) while maintaining an active TS/SCI.
- Test utilized CodeceptJS with JavaScript, HTML, and CSS.

Projects

Operational Amplifier Design

- Designed and optimized MOSFET wide output swing operational amplifier, achieving target specifications, utilizing 0.35 μ m process technology.
- Discussed and researched effective PD techniques for analog IC design.

Design of a Simple CPU

- Designed an 8-bit CPU equipped with 32 bytes of RAM and 3 different op codes
- Implemented on the Intel DE-10 Lite FPGA device and programmed in VHDL

CMOS Logic Optimization

- Calculated the optimal MOSFET width and length for a given CMOS logic network
- CMOS logic network was designed and simulated in LTSpice

Skills

- **Hardware:** Intel DE-10 Lite (FPGA), Cortex-M4, Arduino, Raspberry Pi, Various ICs
- **Languages:** Python, C/C++, VHDL, TCL, Java, x86 and ARM Assembly, MATLAB
- **Software:** PrimeTime, Innovus, KiCad, LTSpice, Xcode, Quartus Prime

Education

West Virginia University Morgantown, WV
M.S. in Electrical Engineering Aug. 2024 – May 2025 (exptd.)

West Virginia University Morgantown, WV
B.S. in Computer Engineering – CGPA: 3.96 Aug. 2020 – May 2024

- Minor in Mathematics
- Electronics Team Lead for WVU University Rover Challenge Team
- President of WVU Alpine Ski Racing Team